



When does AI pay off? AI-adoption intensity, complementary investments, and R&D strategy

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ARTICLE INFO

Keywords:

Artificial intelligence
High-tech ventures
Firm performance
Complementary investment
R&D strategy

ABSTRACT

This paper examines how high-tech venture performance varies with AI-adoption intensity. We find that firm revenue increases only after sufficient investment in AI, and the benefits of AI adoption are greater at firms that also invest in complementary technologies and pursue internal R&D strategy. Specifically, AI adoption at low levels does not suggest significant revenue growth, but, as the intensity of AI adoption increases revenue growth occurs. We find that such performance gains from adoption is larger among firms that invest in complementary technologies such as cloud computing and database systems. Moreover, the positive relationship between AI adoption intensity and revenue growth is stronger among firms that pursue a more exclusive R&D strategy specific to the venture.

1. Introduction

Artificial intelligence (AI) is increasingly being adopted in firms around the world, with its prevalence only expected to increase further. In fact, AI is widely accepted as the newest general-purpose technology (GPT) (Goldfarb et al., 2020; Trajtenberg, 2019), following major technological innovations such as information technology (IT), computers, and electricity. Since AI was first established in computer science in the mid-twentieth century, the development and application of AI have rapidly expanded only in the past decade. Massive investments in AI have occurred recently, and organizations are in the midst of adopting and exploring the best ways to use AI for their purposes. Thus, AI has the potential to change the way we innovate, do business, and organize and to affect all aspects of the economy (Cockburn et al., 2019).

An increasing number of papers examine how AI could affect the economy (Acemoglu et al., 2020; Brynjolfsson et al., 2019, 2021; Chalmers et al., 2020; Farboodi et al., 2019; Mihet and Philippon, 2019). However, the lack of data is a major constraint in the literature and hence many studies have focused on the conceptual implications of AI (Brynjolfsson and Mitchell, 2017; Seamans and Raj, 2018). The few empirical papers on the topic have generally focused on three research streams: 1) the effect of AI on the labor market (Alekseeva et al., 2021; Webb, 2020), 2) whether AI has the features of a general purpose

technology (Goldfarb et al., 2020), and 3) AI adoption and firm or industry level performance of listed firms (Seamans and Raj, 2018). Despite the importance of new ventures for economic growth (Lee, 2018), and technological innovation for entrepreneurial opportunities (Eckhardt and Shane, 2011), we still know surprisingly little about whether AI has made high-tech ventures more productive and, if so, when or why. Furthermore, given that complementary assets are critical to adopting and leveraging new technologies (Rothaermel, 2001; Thomke & Kuemmerle, 2002), less is known about how these complementary assets affect the relationship between the AI adoption of high-tech ventures and their performance (Kim et al., 2021).

This paper seeks to fill this gap by providing empirical evidence on the effects of AI adoption on high-tech venture performance. Specifically, we ask the following research questions: Is AI adoption positively associated with high-tech venture performance? If so, how do complementary technologies and research and development (R&D) strategy moderate this relationship? To answer these questions, we first theorize how AI adoption could improve firm performance, and we examine the relationship between the intensity of AI adoption in high-tech ventures and their performance. Given that complementary investments and organizational changes have been important in new technology's successful contribution to firm performance, complementary technologies and R&D strategy will likely be important for firms to reap the benefits

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<https://doi.org/10.1016/j.technovation.2022.102590>

Received 14 November 2021; Received in revised form 25 May 2022; Accepted 27 June 2022

Available online 4 July 2022

0166-4972/© 2022 Published by Elsevier Ltd.